SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

CONCEPT MAPPING

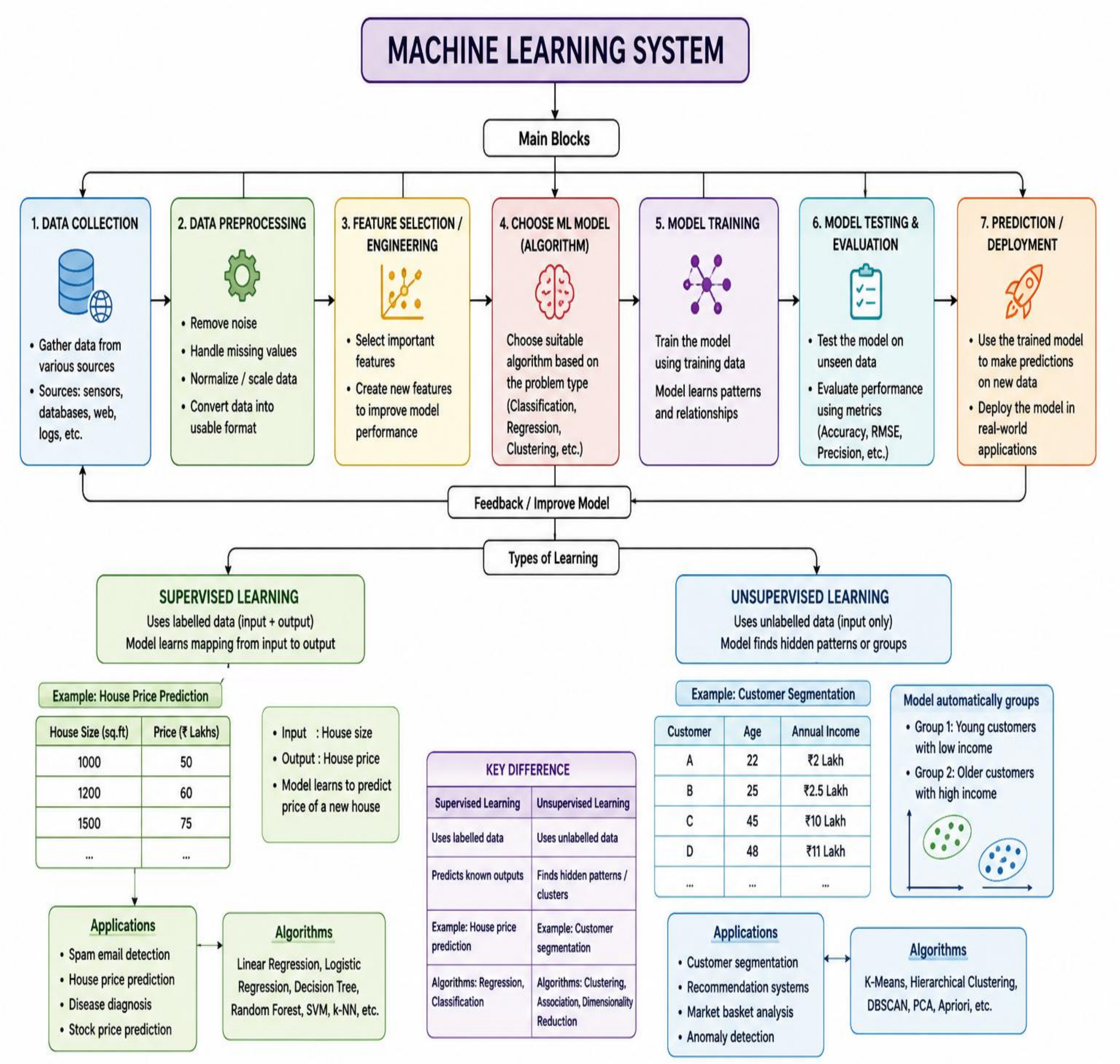
Code of Conduct:

I Nirmal Kumar T certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

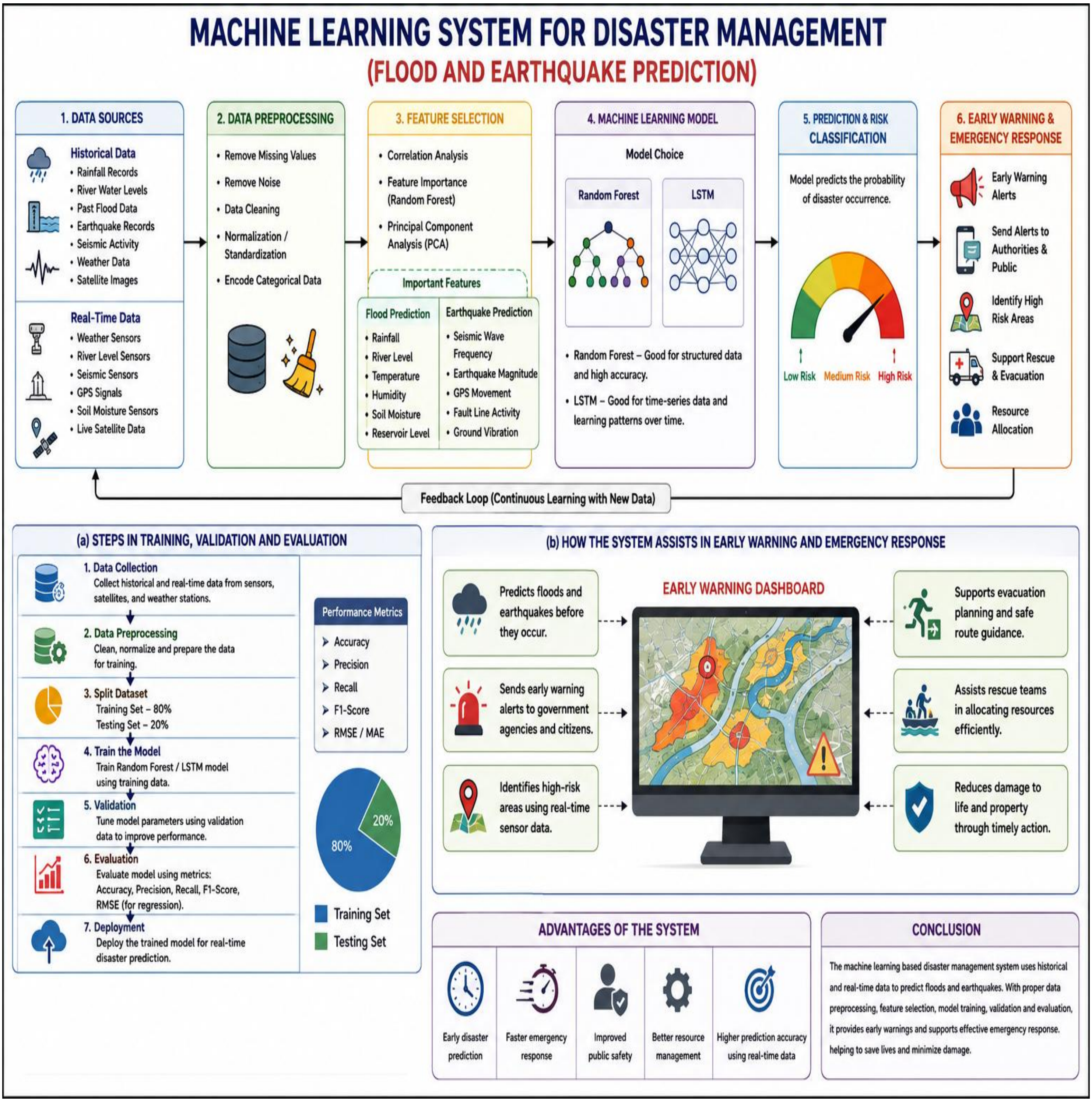
Signature of the student:

Nirmal Kumar T

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.



2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.
(a) Describe the steps involved in training, validating, and evaluating the model.
(b) Explain how the system can assist in early warning and emergency response.



3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

Find-S Algorithm – Play Tennis Dataset

Rule: Start with the most specific hypothesis. Consider only positive examples (Yes). Ignore all negative examples (No). Generalize the hypothesis only when necessary.

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

Step-by-Step Hypothesis Updates (Find-S Algorithm)

Step	Training Example (Only Positive)	Example Values	Current Hypothesis (H)	Explanation
0 (Initial)	–	–	$H_0 = \langle Ø, Ø, Ø, Ø \rangle$	Most specific hypothesis
1	Example 3 (Yes)	⟨Overcast, Hot, High, Weak⟩	$H_1 = \langle$ Overcast, Hot, High, Weak $\rangle$	First positive example becomes hypothesis
2	Example 4 (Yes)	⟨Rainy, Mild, High, Weak⟩	$H_2 = \langle$?, ?, High, Weak $\rangle$	Outlook: Overcast ≠ Rainy → ? Temperature: Hot ≠ Mild → ? Humidity: High = High → High Wind: Weak = Weak → Weak
3	Example 5 (Yes)	⟨Rainy, Cool, Normal, Weak⟩	$H_3 = \langle$?, ?, ?, Weak $\rangle$	Outlook: ? → ? Temperature: ? → ? Humidity: High ≠ Normal → ? Wind: Weak = Weak → Weak
4	Example 7 (Yes)	⟨Overcast, Mild, Normal, Strong⟩	$H_4 = \langle$?, ?, ?, ? $\rangle$	Outlook: ? → ? Temperature: ? → ? Humidity: ? → ? Wind: Weak ≠ Strong → ?
5	Examples 1, 2, 6, 8 (All No)	–	No Change (Still $H_4 = \langle$?, ?, ?, ? $\rangle$)	Negative examples are ignored

Final Maximally Specific Hypothesis

$$H = \langle ?, ?, ?, ? \rangle$$

? = Any value (generalized)

Explanation:
All attributes are generalized to "?" because the positive examples have different values for all attributes. Therefore, no single attribute can uniquely describe all positive examples.

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example.

(5 Marks)

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Warm Rainy		Normal	Strong	No

Candidate Elimination Algorithm – Enjoy Sport Dataset

Dataset

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

Candidate Elimination Algorithm

- S = Set of maximally specific hypotheses (specific boundary)
- G = Set of maximally general hypotheses (general boundary)
- S starts with the most specific hypothesis.
- G starts with the most general hypothesis.
- Update S and G for each training example.
- The version space is all hypotheses h such that $S \le h \le G$ (S is more specific than G).

Notation

Ø : most specific value (matches only that value)

? : any value (most general value)

Step	Training Example	Enjoy Sport	S Boundary (Specific)	G Boundary (General)
0 (Initial)	–	–	$S_0 = \{ \langle Ø, Ø, Ø, Ø \rangle \}$ (most specific)	$G_0 = \{ \langle ?, ?, ?, ? \rangle \}$ (most general)
1	(Sunny, Warm, Normal, Weak)	Yes	$S_1 = \{ \langle$ Sunny, Warm, Normal, Weak $\rangle \}$	$G_1 = \{ \langle ?, ?, ?, ? \rangle \}$ (unchanged)
First positive example → S becomes that example; G remains the same.				
2	(Sunny, Warm, High, Strong)	Yes	$S_2 = \{ \langle$ Sunny, Warm, ?, ? $\rangle \}$	$G_2 = \{ \langle ?, ?, ?, ? \rangle \}$ (unchanged)
Second positive → generalize S where examples differ (Humidity, Wind).				
3	(Rainy, Cold, High, Weak)	No	$S_3 = \{ \langle$ Sunny, Warm, ?, ? $\rangle \}$ (unchanged)	$G_3 = \{ \langle$ Sunny, ?, ?, ? $\rangle$, $\langle$?, Warm, ?, ? $\rangle \}$
Negative example → remove any G that is inconsistent; generalize remaining G minimally to exclude the negative example.				
4	(Sunny, Cold, Normal, Weak)	Yes	$S_4 = \{ \langle$ Sunny, ?, ?, Weak $\rangle \}$	$G_4 = \{ \langle$ Sunny, ?, ?, ? $\rangle \}$ (remove hypothesis $\langle$?, Warm, ?, ? $\rangle$; then minimize to stay $\ge$ S)
Positive example → generalize S to cover it; remove from G any hypothesis that is less general than new S.				
5	(Rainy, Warm, Normal, Strong)	No	$S_5 = \{ \langle$ Sunny, ?, ?, Weak $\rangle \}$ (unchanged)	$G_5 = \{ \langle$ Sunny, ?, ?, ? $\rangle \}$ (unchanged)
Negative example → no change (already excluded by G_4).				

Final Version Space

Specific Boundary (S) = $\{ \langle$ Sunny, ?, ?, Weak $\rangle \}$

General Boundary (G) = $\{ \langle$ Sunny, ?, ?, ? $\rangle \}$

All hypotheses h such that $\langle$ Sunny, ?, ?, Weak $\rangle \le h \le \langle$ Sunny, ?, ?, ? $\rangle$ are consistent with the training data.

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.

(5 Marks)

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

Find-S ALGORITHM – CUSTOMER BUY PRODUCT DATASET

DATASET

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

Find-S Algorithm

- Start with the most specific hypothesis.
- Consider only positive examples (Yes).
- Ignore all negative examples (No).
- Generalize the hypothesis only when necessary.

STEP-BY-STEP HYPOTHESIS UPDATES (CONSIDERING ONLY POSITIVE EXAMPLES)

Step	Training Example	Example Values (Age, Income, Student, Credit Rating)	Positive?	Hypothesis (h) After Update
0 (Initial)	–	–	–	h_0 = < Ø, Ø, Ø, Ø > (most specific hypothesis)
1	Example 1	(Young, High, No, Fair)	No	Ignored (negative example)
2	Example 2	(Young, High, No, Excellent)	No	Ignored (negative example)
3	Example 3	(Middle, High, No, Fair)	Yes	h_1 = < Middle, High, No, Fair > (first positive example becomes hypothesis)
4	Example 4	(Senior, Medium, No, Fair)	Yes	Compare with h_1 = < Middle, High, No, Fair > Age: Middle vs Senior → ? Income: High vs Medium → ? Student: No vs No → No Credit Rating: Fair vs Fair → Fair h_2 = < ?, ?, No, Fair >

Explanation of Generalization

- Age differs (Middle vs Senior) → generalize to ?
- Income differs (High vs Medium) → generalize to ?
- Student same (No vs No) → keep No
- Credit Rating same (Fair vs Fair) → keep Fair

FINAL MAXIMALLY SPECIFIC HYPOTHESIS

h_{final} = < ?, ?, No, Fair >

Meaning:

- Age can be any value
- Income can be any value
- Student must be No
- Credit Rating must be Fair

Thus, the maximally specific hypothesis obtained using the Find-S algorithm is: < ?, ?, No, Fair >

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand